Project Report:

Endometriosis

STATISTICAL ANALYSES AND VISUALIZATIONS IN R:I

VALERIA ANGELINO

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1. Introduction

According to the WHO (2023), endometriosis is a chronic disease that affects around 190 million of women and girls globally. According to the SCI International Hospital, endometriosis is a condition where the endometrium - the tissue which resembles the uterus' lining -, grows outside of the uterus (SCI, 2025), leading to inflammation and scar tissue forming (WHO, 2025). Endometriosis is associated with severe, life-impacting pain periods, chronic pelvic pain, abdominal bloating, nausea, fatigue, and sometimes depression, anxiety, and infertility. Despite this, there is little research on endometriosis, and the precise cause of it and any possible cures are unknown (WHO, 2023). This paper aims to examine how variables like diagnosis, chronic pain level, age, body mass index (BMI), and infertility relate to endometriosis, highlighting relevant patterns and trends.

Endometriosis can decrease quality of life; for instance, chronic pain can be very strong that prevents people from going to work or school (WHO, 2023). This prompted the question of whether, by analyzing a dataset, one could identify if women with a confirmed diagnosis of endometriosis would report higher chronic pain levels. This leads to the first research question of this paper:

RQ1: Do women with an endometriosis diagnosis report a higher average chronic pain compared to those without a diagnosis?

Further, according to the SCI International Hospital (2025), the impact of age can vary among people, with some experiencing a reduction in symptoms, and others worsening. Mainly, the immune system naturally weakens with age, which may reduce the body's ability to suppress the growth of endometrial tissue, potentially resulting in the recurrence of symptoms (SCI, 2025). This made me wonder if there is a correlation between age and chronic pain level. Therefore, the second research question of this paper is:

RQ2: Is there a significant correlation between age and chronic pain level in individuals with endometriosis?

As per a research paper by Tang et al. (2020), body mass index (BMI) is thought to be a potential risk factor in the development of endometriosis. However, Tang et al. (2020) note that the

association between BMI and endometriosis remains controversial. Therefore, this paper aims to explore if different BMI's are associated with different chronic pain levels by addressing the following research question:

RQ3: Is there a significant difference in chronic pain levels across different BMI categories in women with endometriosis?

Finally, according to Cleveland Clinic (2024), some people with endometriosis have trouble getting pregnant due to scarring and fallopian tube blockage. This leads to the final question of this paper:

RQ4: Is there a significant association between being diagnosed with endometriosis and experiencing infertility?

To adress the four research questions, I used a synthetic dataset designed to simulate real-world scenarios by including common symptoms associated with endometriosis. Appropriate statistical analyses were then conducted in RStudio, including a t-test of independence, a correlation analysis, a Kruskal-Wallis test, and a Chi-square test.

The goal of this paper is to use statistical methods to explore potential trends and relationships related to this condition. Given the significant impact of endometriosis in millions of women, it is important to get a better understanding of the condition, the extent of its impact, and how it interacts with other factors. Although this paper is based on synthetic data, the analyses conducted can still contribute to a deeper understanding of endometriosis, raising awareness, and inspiring future research on endometriosis.

2. Data & Methods

The dataset used in this paper was extracted from Kaggle. According to the dataset author, the dataset is based on synthetic data aimed at simulating a real scenario by including common symptoms associated with endometriosis. Each research question is answered via a statistical analysis performed in RStudio for which the code used has been included as an Appendix. ChatGPT Open AI was used to correct code syntax and errors.

2.1. RQ1: Do women with an endometriosis diagnosis report a higher average chronic pain compared to those without a diagnosis?

This question dives into the difference in the means of a continuous variable (*chronic_pain_level*) in two independent groups: either a woman has an endometriosis diagnosis, or they do not. According to Salkind (2017), a sample t-test would be appropriate to answer this question. Firstly, I obtained a summary and measures of central tendency to get a better insight into the data. Next, I had to find out the distribution of the numerical variable *chronic_pain_level* to determine if a t-test is suitable. For that, I created a histogram of the distribution of *chronic_pain_level*, shown in Figure 1. From Figure 1 I can see that my variable *chronic_pain* meets the normality assumption for a t-test. The null hypothesis in this case is that there is no difference in the means, and the alternative hypothesis would reflect a difference in the means of chronic pain between the two groups.

Before I ran the t-test, I looked into the measures of central tendency of the variable's *diagnosis* and *chronic_pain_level* to get an insight on the data. The next step was to run a sample t-test of independence using Rcmdr. The results have been gathered into Table 3. Finally, for a better visualization, a violin plot with boxplots (Figure 2) was created. The overall results are then discussed in the Results and Discussion section.

2.2. RQ2: Is there a correlation between age and chronic pain level in individuals with endometriosis? (correlation)

To answer this question, we have two continous variables *Age* and *Chronic Pain Level.* To find a correlation between them, the fitting tests would be a Pearson's Product Movement Correlation or Spearman's Rank Correlation depending on the normal distribution of the data. Therefore, the first step is to find wether the two variables *Age* and *Chronic Pain Level* are normally distributed. The variables' normality will be tested using histograms and a Shapiro-Wilk normality test. This test is recommended for normality test, as it is based on the correlation between the data and the corresponding normal scores (Kassambara, 2020).

Given that the results of the Shapiro-Wilk test (Table 4) show that none of the variables follow a normal distribution, the best test to find a correlation would be a nonparametric Spearman's Rank Correlation test. A Spearman's correlation test will allow us to assess if there is a monotonic

relationship between *Age* and *Chronic Pain Level* – a monotonic function refers to a relationship where one variable consistently increases or decreases as the independent variable increases (Statstutor, n/d). The results of the Spearman's test have been shown in Table 5. For a visualization of the data, I created a scatter plot with a smooth curve (LOESS) that adapts to different parts of the data revealing further patterns, shown as Figure 4.

2.3. RQ3. Is there a significant difference in chronic pain levels across different BMI categories in women with endometriosis?

This question considers the difference between multiple groups across the independent *BMI* variable; underweight, normal, and Overweight. On the other hand, the dependent variable is *chronic_pain_level*, a numerical continuous variable.

Given our variables, a one-way ANOVA is recommended (Salkind, 2017) as long as the assumptions of homoscedasticity and normality in the residuals are met. Firstly, I look at the diagnostic plots of the one-way ANOVA to get an insight into the normality and homoscedasticity assumptions, shown as Figure 5. For a better insight, to assess the assumption of normality, I ran a Shapiro-Wik test, for which the results are shown in Table 6.

Given that the assumptions of equal variance and normality of the one-way ANOVA are not met, a non-parametric Kruskal-Wallis test will be performed. This test will allow us to compare the overall chronic pain level difference between the different BMI groups. The null hypothesis of the Kruskal-Wallis test H0: The three BMI groups have the same median chronic pain level, and the alternative hypothesis H1: At least one BMI group has a different median of chronic pain level. The results are shown in Table 7. Given that the results of the Kruskal-Wallis test did not indicate a significant difference among the different BMI groups, no post-hoc test was needed.

To better visualize the distribution of chronic pain levels across BMI categories, I created a violin plot with an overlaid boxplot: Figure 6.

2.4. RQ4: Is there a significant association between being diagnosed with endometriosis and experiencing infertility?

This question considers count data and the association between two variables: *Diagnosis* and *Infertility*, therefore, it can be answered using a chi-square test of independence. First, I began by

creating a contingency table shown as Table 8. Once the contingency table was created, I moved onto performing a Chi-squared test of goodness of fit. The null hypothesis for the Chi-square value was H0: There is no association between endometriosis and infertility, whilst the alternative hypothesis reflected H1: There is a significant association between endometriosis and infertility. The results of the Chi-square test were tabulated in Table 9. Moreover, I created a bar chart shown as Figure 7 for a more visual interpretation.

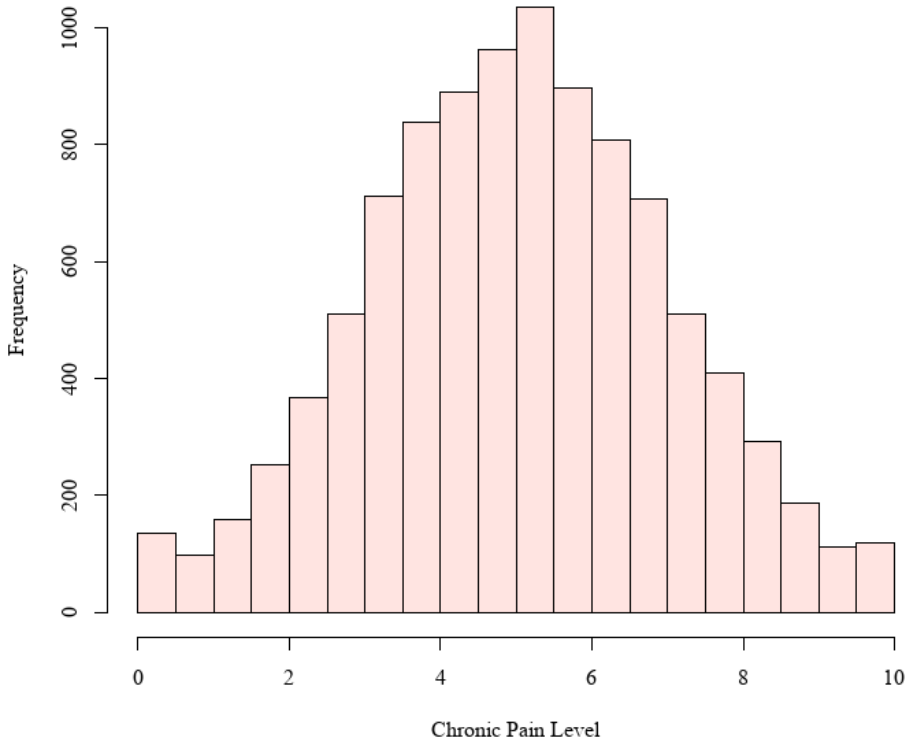
3. Results

3.1. RQ1. Endometriosis and Chronic Pain

Table 1. Summary of Diagnosis	
No Endometriosis	Endometriosis
5921	4079

Table 2. Summary of Chronic Pain Level			
Min	Median	Mean	Max
0.000	5.036	5.031	10

Figure 1. Chronic Pain Level Distribution



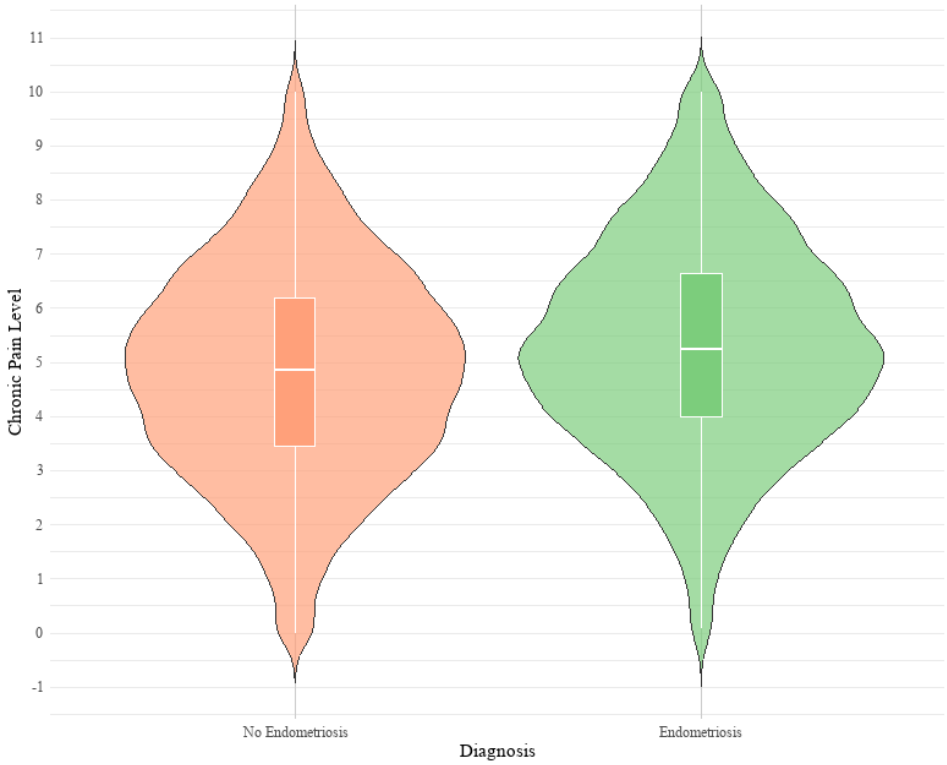
From Figure 1, we can see that *chronic pain level* follows a bell shape, which suggests that the data is normally distributed.

Table 3. Independent Sample t-test Results	
t-value	-11.792
df	8798.5
p-value	<2.2e-16
Chronic Pain Mean in group with No Endometriosis	4.837972
Chronic Pain Mean in group with Endometriosis	5.310260

As the p-value<0.05, the null hypothesis is rejected and statistically significant difference in chronic pain level between women with or without an endometriosis diagnosis is concluded. Similarly, the absolute t-value (-11.792) is very large, suggesting a strong difference between both diagnosis groups. In other words, the t-value shows that the mean chronic pain level for women

with an endometriosis diagnosis is higher than those without one, as shown in Table 3. These results suggest that there is a highly significant difference in chronic pain levels between women with and without an endometriosis diagnosis, where women with a diagnosis tend to experience higher chronic pain.

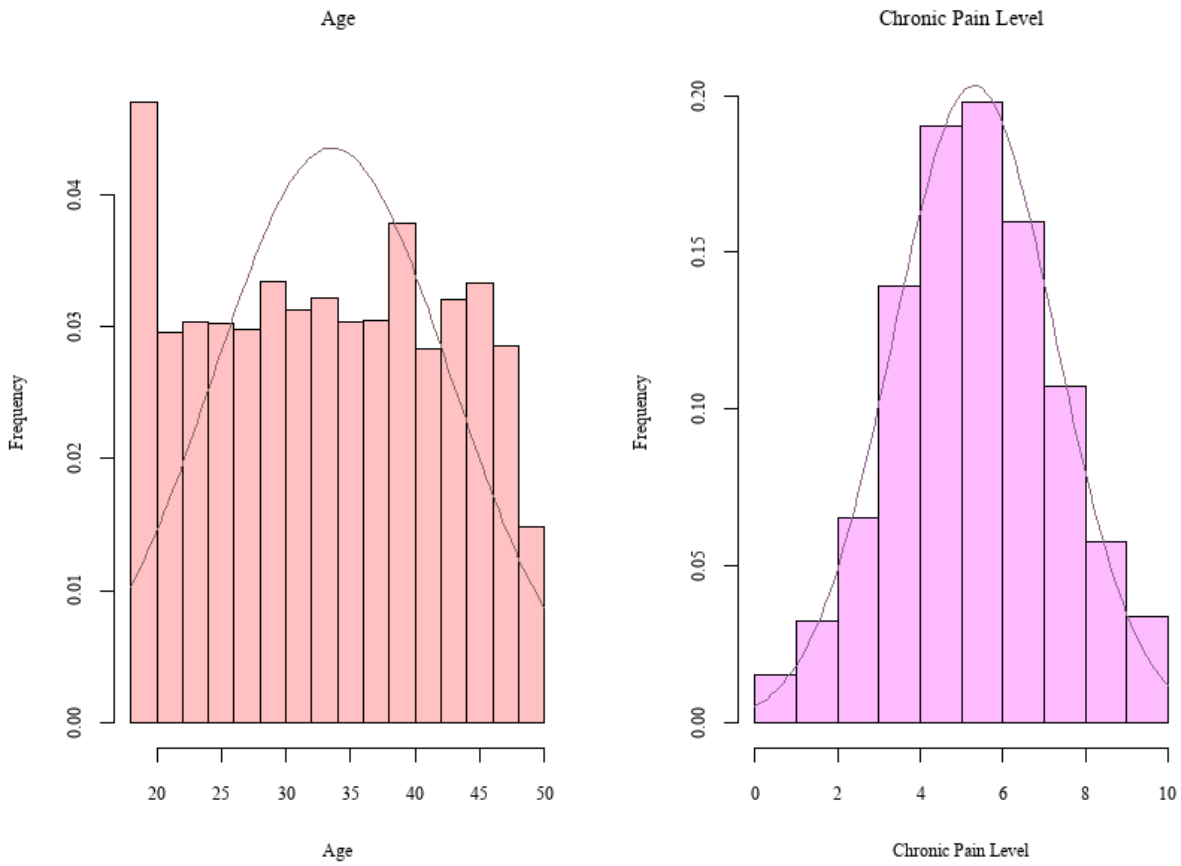
Figure 2. Chronic Pain Levels by Diagnosis



In Figure 2 we can visually see the difference in chronic pain level across the two diagnosis groups: no endometriosis and endometriosis. We can also see that more women with endometriosis experience higher chronic pain level thank those without endometriosis. We can also see that lower chronic pain level is more frequent for women with no endometriosis than those with. Finally, by the violin shape and the built-in boxplot we can see that women with endometriosis experience a higher level of chronic pain than those without endometriosis.

3.2 RQ2. Age and Chronic Pain Level

Figure 3. Distribution of Age and Chronic Pain Level for Women with Endometriosis)



The histograms in Figure 3 show that our variable *Age* does not follow a normal distribution, whilst *Chronic Pain Level* seems to follow a bell-shaped distribution. For a better insight into their normality, we can look at the results of the Shapiro-Wilk test shown in Table 4. The null hypothesis for the Shapiro-Wilk test states that H0: the sample follows a normal distribution, whilst the alternative hypothesis H1: The sample does not follow a normal distribution.

Table 4. Shapiro-Wilk Normality Test		
	Age	Chronic Pain Level
P-value	2.2e-16	1.931e-07

The P-values for both variables are smaller than 0.05, therefore, we have to reject the null hypothesis (H0). In other words, despite a bell-shaped histogram, the Shapiro-Wilk test points that both variables are not normally distributed.

Table 5. Spearman's Rank Correlation Test Results	
p-val	rho
0.3217	-0.0155

Firstly, the p-value is higher than 0.05, suggesting that there is not a statistically significant association between the two variables. Further, the rho value of -0.0155 is very close to 0, which indicates no monotonic relationship. This means that there is no statistically significant correlation between age and chronic pain level.

Figure 4. Age vs. Chronic Pain Level

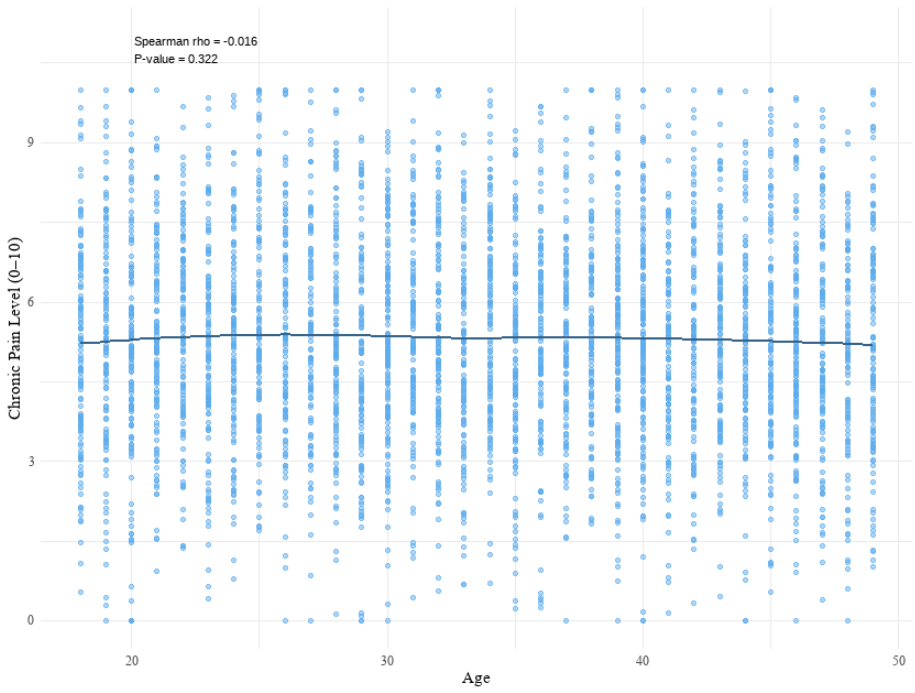


Figure 4 shows a scatter plot of the relationship between *Age* and *Chronic Pain* Level with a smooth curve (LOESS line). We can now visually see that there is no statistical relationship between age and chronic pain level, as chronic pain level is evenly distributed across different ages with no clear pattern or trend.

3.3 RQ3.Chronic Pain Levels and BMI

Figure 5. Diagnostic Plots of One-Way ANOVA

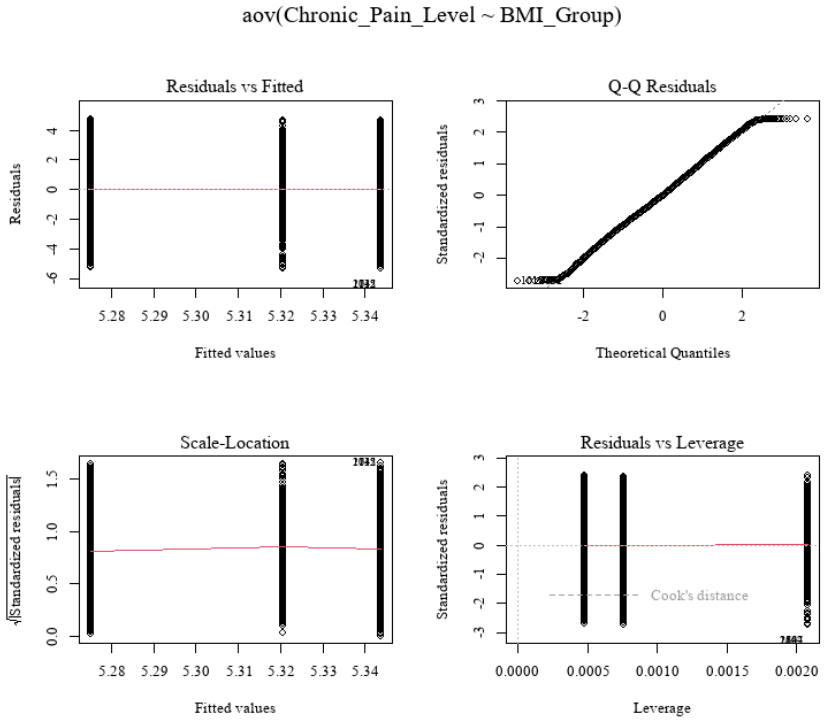


Table 6. Shapiro-Wilk Normality Test	
P-value	7.696e-07

In the Residuals vs Fitted plot in Figure 5, we can see that the residuals look randomly scattered along the horizontal red line, and there is no pattern in the residuals, indicating homoscedasticity. Figure 5 also shows that there is a skewness in the distribution of the residuals (Q-Q Residuals), which could indicate a non-normal distribution. For a clearer answer, the results of the Shapiro-Wilk Normality test shown in Table 6 show a P-value of 7.696e-07, which is less than the significance level of 0.05. This means that we reject the null hypothesis (H0) of normality, therefore the residuals of the one-way ANOVA are not normally distributed. Given that the assumptions for a one-way ANOVA are not met, a Kruskal-Wallis test will be performed instead.

Table 7. Kruskal-Wallis Test		
Chi-squared	df	P-value

1.4746	2	0.4784

Table 7 shows the results of the Kruskal-Wallis test, showing a p-value of 0.4784. Given that the P-value is higher than the 0.05 significance level, we accept the null hypothesis H0: The three BMI groups have the same median chronic pain level in women with endometriosis.

Figure 6. Chronic Pain Levels by BMI Group

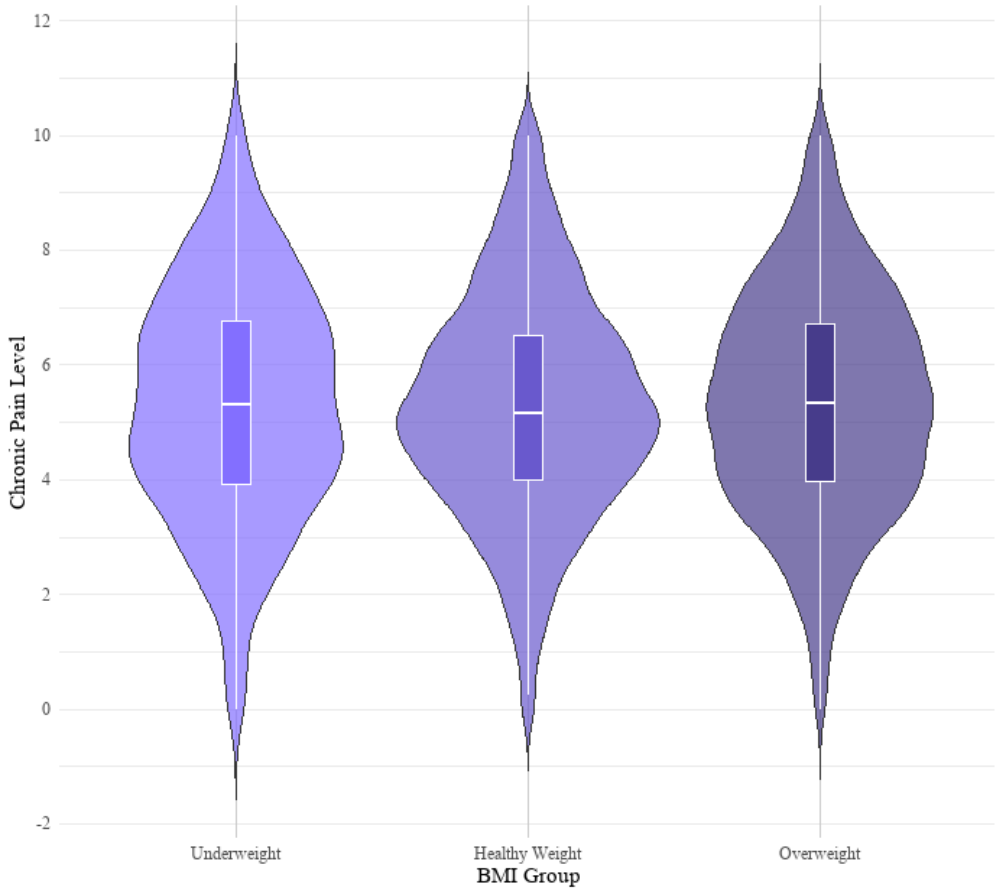


Figure 6 shows the variation and tendencies of chronic pain across the three different BMI groups. We can see that women in the underweight category seem to experience both lower and higher levels of pain at a higher frequency compared to the healthy weight and overweight groups. However, the median chronic pain is around the same level for all three groups.

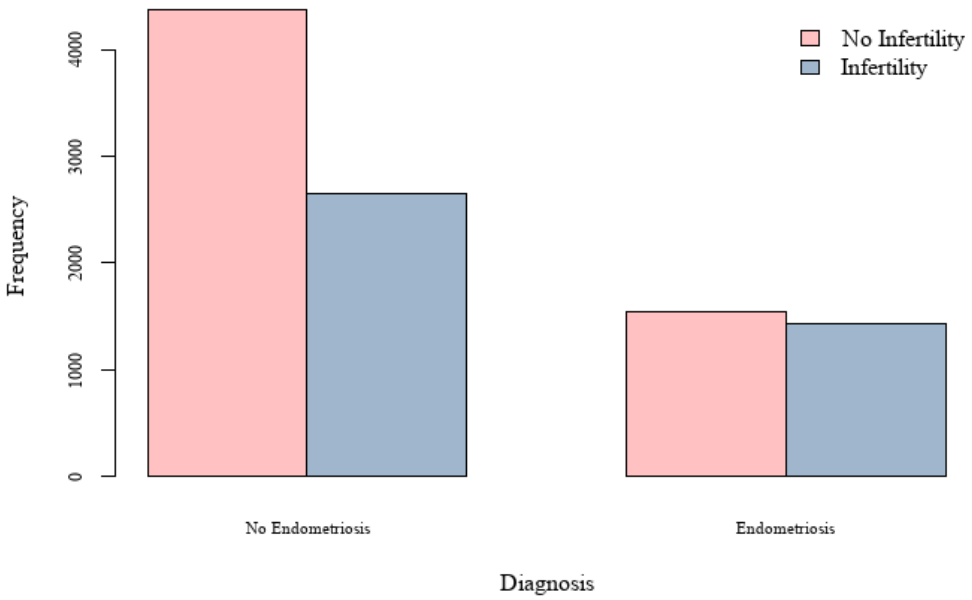
3.4 RQ4. Endometriosis and Infertility

Table 8. Contingency Table of Diagnosis & Infertility		
	Does Not Experience Infertility	Experiences Infertility
No Endometriosis	4371	1550
Endometriosis	2646	1433

Table 9. Chi-square test of Diagnosis & Infertility		
X-Squared	df	*P-value*
92.063	1	2.2 e-16

In Table 9, we can see that the P-value of the Chi-square tests is smaller than the 0.05 significance level. Therefore, we can reject the null hypothesis and accept the alternative hypotehsis H1: There is a significant association between endometriosis and infertility.

Figure 7. Endometriosis Diagnosis and Infertility



From Figure 7, we get a better visualization of the data. We can see that although there is a higher frequency of women without an endometriosis diagnosis that experience infertility, women that have an endometriosis diagnosis appear to experience infertility more commonly. Infertility seems to be an issue that affects almost half of women that have an endometriosis diagnosis, as we see similar frequencies. Thus, we can see that there is a significant association between a positive endometriosis diagnosis and infertility.

4. Discussion & Conclusion

From the results presented in the previous section provide a deeper insight on the impact and implications of endometriosis. Firstly, the first RQ aimed to answer if *women with endometriosis reported a higher average chronic pain level than those without a diagnosis*. The results of the t-test showed that there is a statistically significant difference in the chronic pain level between women with and without an endometriosis diagnosis. This difference could be visualized in Figure 2, where we could see a higher frequency in higher chronic pain levels for women with an endometriosis diagnosis. Therefore, we can say that women with endometriosis do report a higher chronic pain level than those without an endometriosis diagnosis.

RQ2 provided an insight into the chronic pain levels among women with endometriosis, and whether age was correlated with chronic pain. The results indicated no statistically significant correlation between age and chronic pain level, given the P-value of 0.3217. This is significant given that SCI International Hospital (2025) suggested that due to the age-related deterioration of the body, symptoms such as chronic pain level would be more prevalent. The lack of correlation between age and chronic pain level highlights the need for further research as there could be other factors (beyond age) that may be correlated with a higher chronic pain level among women with endometriosis.

The association between body mass index (BMI) and chronic pain levels in women with endometriosis was covered by RQ3. This indirectly adds to the research by Tang et al (2020), who pointed to BMI as a potential factor in the development of endometriosis. The results showed that there is no significant association between BMI and experiencing higher or lower chronic pain levels. The visualization of the pain differences between BMIs in Figure 6 still show different

distributions in the reported pain levels. Therefore, more research in the impact of BMI in the symptoms of endometriosis is needed.

RQ4 allowed us to find a significant association between endometriosis and infertility. The Chi-squared test ran and the visualization in Figure 7 showed that women diagnosed with endometriosis experienced infertility at a higher frequency than those without a diagnosis. This finding aligns with Cleveland Clinic's findings, suggesting that endometriosis can affect fertility. This association calls for future research to explore factors that could mitigate this association, as well as potential interventions to reduce infertility among women with endometriosis.

Finally, this paper highlighted potential trends and relationships related to endometriosis by answering the four research questions:

RQ1: Do women with an endometriosis diagnosis report a higher average chronic pain compared to those without a diagnosis?

RQ2: Is there a significant correlation between age and chronic pain level in individuals with endometriosis?

RQ3: Is there a significant difference in chronic pain levels across different BMI categories in women with endometriosis?

RQ4: Is there a significant association between being diagnosed with endometriosis and experiencing infertility?

Despite the findings of this paper, further research is needed to fully understand endometriosis and the extent of its impact on the lives of millions of women.

5. References

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Statstutor (n/d). Spearman's Correlation [pdf], Available online: https://www.statstutor.ac.uk/resources/uploaded/spearmans.pdf

6. Appendix

```
library(dplyr)
library(ggplot2)
library(scales)
library(corrplot)
library(flextable)
library(faraway)
library(car)

df <- read.csv ("C:/Users/angel/OneDrive/Desktop/School/R Course/Endometriosis
Project/structured_endometriosis_data.csv")

View(df)

windowsFonts(TimesNewRoman = windowsFont("Times New Roman"))
par(family = "Times New Roman")

font_add("Times New Roman", regular = "C:/Windows/Fonts/times.ttf")
showtext_auto()

# Changing binary data to named binary data

df <- df %>%
  mutate(
    Menstrual_Irregularity = factor(Menstrual_Irregularity, labels = c("No",
"Yes")),
    Hormone_Level_Abnormality = factor(Hormone_Level_Abnormality, labels =
c("Normal", "Abnormal")),
    Infertility = factor(Infertility, labels = c("No", "Yes")),
    Diagnosis = factor(Diagnosis, labels = c("No Endometriosis", "Endometriosis"))
  )

summary(df)
```

6.1. RQ1

```r
# RQ1 - diagnosis and chronic pain t-test

summary(df$Diagnosis)
summary(df$Chronic_Pain_Level)

hist(df$Chronic_Pain_Level,
     col = "mistyrose",
     border = "black",
     main = "Figure 1. Chronic Pain Level Distribution",
     xlab = "Chronic Pain Level",
     ylab= "Frequency"
)

t.test(Chronic_Pain_Level~Diagnosis, alternative='two.sided',
       conf.level=.95, var.equal=FALSE, data=df)

# Visualization

ggplot(df, aes(x = Diagnosis, y = Chronic_Pain_Level, fill = Diagnosis)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.1, color = "white", outlier.shape = NA) +
  theme_minimal() +
  labs(
    title = "Figure 2. Chronic Pain Levels by Diagnosis",
    x = "Diagnosis",
    y = "Chronic Pain Level"
  ) +
  scale_fill_manual(values = c("lightsalmon", "palegreen3")) +
  theme(
    text = element_text(family = "Times New Roman"),
    legend.position = "none",
    panel.grid.major.x = element_line(color = "gray80"),
    panel.grid.minor.x = element_line(color = "gray90")
  ) +
  scale_y_continuous(
    breaks = scales::pretty_breaks(n = 10)
  )
```

6.2. RQ2

```
# RQ2 - Correlation Age & Chronic Pain

library(dplyr)
just_endo <- df %>% filter(Diagnosis == 'Endometriosis')

par(mfrow=c(1,2), oma = c(0, 0, 1.5, 0), family = "Times New Roman")

hist(just_endo$Age,
     col = "rosybrown1",
     border = "black",
     main = "Age",
     xlab = "Age",
     ylab= "Frequency",
     cex.main = 0.9,
     font.main = 1,
     cex.axis = 0.75,
     cex.lab = 0.75,
     prob = TRUE
)
curve(dnorm(x, mean = mean(just_endo$Age, na.rm = TRUE),
            sd = sd(just_endo$Age, na.rm = TRUE)),
      col = "rosybrown4", lwd = 1, add = TRUE)

hist(just_endo$Chronic_Pain_Level,
     col = "plum1",
     border = "black",
     main = "Chronic Pain Level",
     xlab = "Chronic Pain Level",
     ylab= "Frequency",
     cex.main = 0.9,
     font.main = 1,
     cex.axis = 0.75,
     cex.lab = 0.75,
     prob = TRUE
)
curve(dnorm(x, mean = mean(just_endo$Chronic_Pain_Level, na.rm = TRUE),
            sd = sd(just_endo$Chronic_Pain_Level, na.rm = TRUE)),
      col = "plum4", lwd = 1, add = TRUE)

mtext("Figure 3. Distribution of Age and Chronic Pain Level for Women with
Endometriosis)",
      outer = TRUE, cex = 1)

    # Shapiro-Wilk Test

shapiro.test(just_endo$Age)
shapiro.test(just_endo$Chronic_Pain_Level)
```

```r
 # Correlation

library(ggplot2)

cor_result <- cor.test(just_endo$Age, just_endo$Chronic_Pain_Level, method =
"spearman")
rho <- round(cor_result$estimate, 3)
pval <- signif(cor_result$p.value, 3)

ggplot(just_endo, aes(x = Age, y = Chronic_Pain_Level)) +
  geom_point(alpha = 0.5, color = "steelblue2") +
  geom_smooth(method = "loess", color = "steelblue4", se = FALSE) +
  labs(title = "Figure 4. Age vs. Chronic Pain Level",
       x = "Age", y = "Chronic Pain Level (0-10)") +
  annotate("text",
           x = max(just_endo$Age) * 0.41,
           y = max(just_endo$Chronic_Pain_Level)*1.1,
           label = paste0("Spearman rho = ", rho, "\nP-value = ", pval),
           hjust = 0, vjust = 1, size = 3) +
  theme_minimal() +
  theme(
    text = element_text(family = "Times New Roman")
  ) +
  coord_cartesian(clip = "off")
```

6.3. RQ3

```
# RQ3 - Endometriosis and BMI

just_endo <- na.omit(just_endo)

just_endo$BMI_Group <- cut(just_endo$BMI,
                           breaks = c(-Inf, 18.5, 25, 30),
                           labels = c("Underweight", "Healthy Weight",
"Overweight"),
                           right = FALSE)

anova_model <- aov(Chronic_Pain_Level ~ BMI_Group, data = just_endo)

          # Checking assumptions

shapiro.test(residuals(anova_model))

par(mfrow = c(2, 2))
plot(anova_model)

          # Kruskall

kruskal.test(Chronic_Pain_Level ~ BMI_Group, data = just_endo)

          # Visualization

ggplot(just_endo, aes(x = BMI_Group, y = Chronic_Pain_Level, fill = BMI_Group)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.1, color = "white", outlier.shape = NA) +
  theme_minimal() +
  labs(
    title = "Figure 6. Chronic Pain Levels by BMI Group",
    x = "BMI Group",
    y = "Chronic Pain Level"
  ) +
  scale_fill_manual(values = c("slateblue1", "slateblue3", "slateblue4")) +
  theme(
    text = element_text(family = "Times New Roman"),
    legend.position = "none",
    panel.grid.major.x = element_line(color = "gray80"),
    panel.grid.minor.x = element_line(color = "gray90")
  ) +
  scale_y_continuous(
    breaks = scales::pretty_breaks(n = 10)
  )
```

6.4. RQ4

```
# RQ4 - diagnosis and infertility

    # Creating a contingency table

Contingency.Table1 <- table( "rows" = df$Diagnosis, "columns" = df$Infertility)

Contingency.Table1

    # Chi square test
Chi.Test1 <- chisq.test(Contingency.Table1)

Chi.Test1

    # VISUALIZATION Barplot
windowsFonts(TimesNewRoman = windowsFont("Times New Roman"))
par(mfrow=c(1,1), family = "Times New Roman")
barplot(Contingency.Table1,
        beside=T,
        col=c("rosybrown1", "slategray3"),
        xlab="Diagnosis",
        ylab="Frequency",
        main="Figure 7. Endometriosis Diagnosis and Infertility",
        legend.text=c("No Infertility", "Infertility"),
        args.legend = list(x="topright", bty="n"),
        cex.axis = 0.75,
        cex.names = 0.7,
        names.arg = c("No Endometriosis", "Endometriosis")
        )
```